MKGW4 Remote Config Tool

User Manual

Version V1.0

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1. Introduction

MOKO provides a local configure tool and remote configure tool for users to test the MKGW4 gateway. The local configure tool is a configuration APP, all configurations are done via Bluetooth. The remote config tool is a computer software, all configurations are done by MQTT commands.

This User Guide was designed to help users to view the BLE scanning data and GNSS data uploaded from the gateway and remotely configure/manage the gateway via MOKO provided computer software.

2. Software setup

Unzip the MKGW4Tools file. Click and open the folder, then you can find the MKGW4Tools as shown below. Double click the tool, it will start running.

名称	修改日期	类型	大小
bin	2024/1/12 19:46	文件夹	
HisData	2024/1/12 19:46	文件夹	
ComponentFactory.Krypton.Toolkit.dll	2023/1/16 15:00	应用程序扩展	2,608 KB
ExcelDataReader.dll	2019/5/2 12:39	应用程序扩展	158 KB
ExcelDataReader.pdb	2019/5/2 12:39	PDB 文件	64 KB
MKGW4Tools	2024/1/11 9:09	应用程序	1,019 KB
MKGW4Tools.exe.config	2023/12/15 14:09	CONFIG 文件	1 KB
MKGW4Tools.pdb	2024/1/8 15:59	PDB 文件	802 KB
MQTTnet.dll	2022/7/12 22:46	应用程序扩展	283 KB
System.Data.SQLite.dll	2016/4/14 11:56	应用程序扩展	1,214 KB

3. Interface layout

3.1 Home page

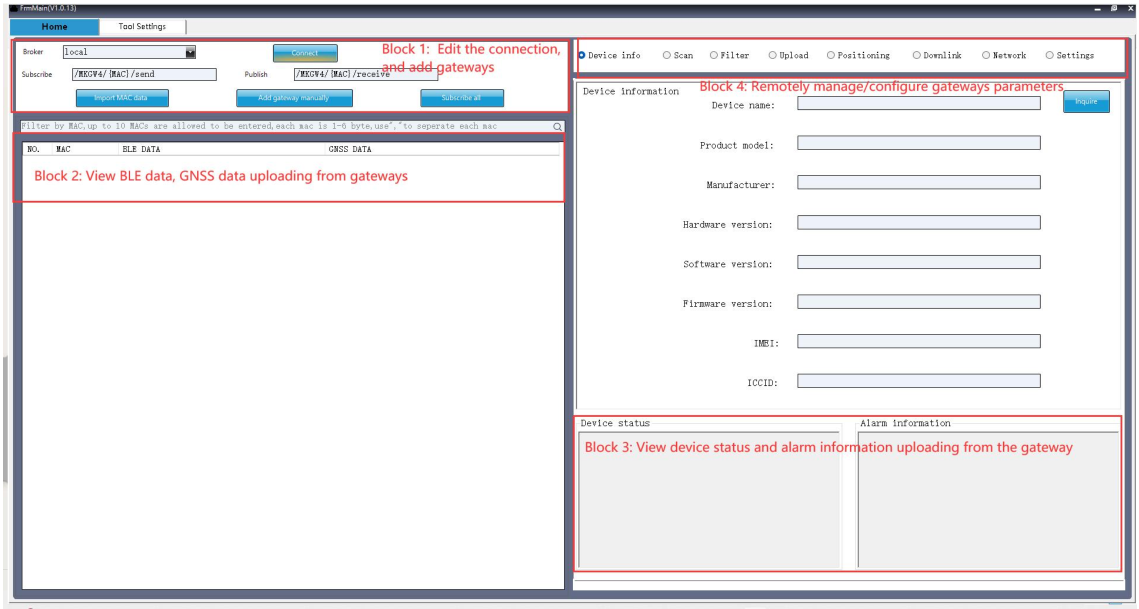
The home page includes four main functions:

Block 1: Edit the connection to the MQTT broker, add gateway

Block 2&3: View gateway uploading data:

- BLE data: The captured BLE data are presented in the left table which corresponds to its device MAC.
- GNSS data:The GNSS data are presented in the left table which corresponds to its device MAC.
- Device status: The device status which includes gateway network status, battery voltage and ACC status are presented in the bottom right corner box.
- Alarm information:The alarm information are presented in the bottom right corner box.

Block 4: Read gateway basic information and configure gateway parameters

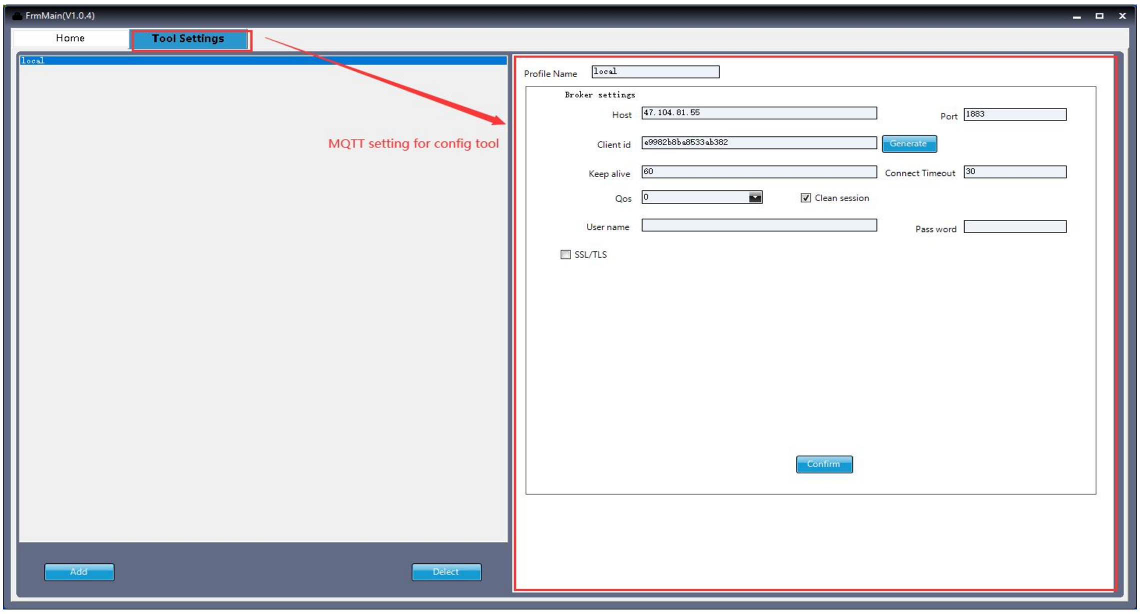


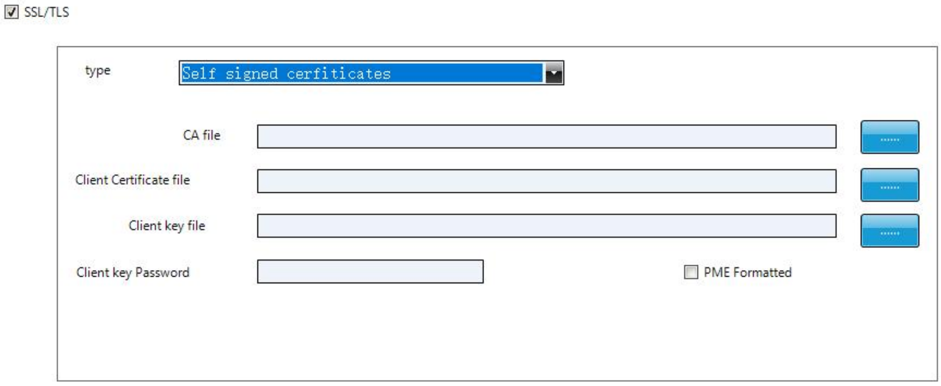
3.2 Tool settings

In the tool settings page you can edit MQTT parameters for this config tool. It allows you to edit and save multiple broker settings.

Click the  button to add new broker settings, we can edit the profile name, and the broker host, port and some other settings. If the MQTT broker requires SSL/TLS encryption, please check the SSL/TLS option and upload corresponding certificate files. The certificate files will be obtained from the local PC. Once all the MQTT settings have been accomplished, click the confirm button to save the new configuration.

The  button is used to remove MQTT broker information from this tool. We can select the broker name in the left panel, click the delete button, then the MQTT broker name will be disappeared from the list and all information of the broker will be deleted.





Note: The MQTT broker applied for this tool should be consistent with the one for the gateways, otherwise the config tool can not communicate with the gateways.

4. Basic function

4.1 Edit connection and add gateways

If the gateways are applied with default MQTT settings (***47.104.81.55:1883***), and their subscription and publish topics are not changed on the configuration app, we can follow the below steps to add the gateways to this tool:

1. Select the local broker, and click 

button. If it connect successfully, the button will change to disconnect.

2. Click 

to upload the gateway mac address from your computer. After that, we will see the gateway mac are listed in the data table.

3. Click 

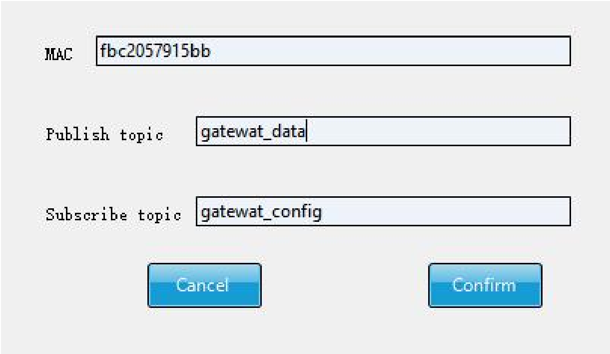
button to make the tool start monitoring the gateway data flow.

If the gateway are applied with custom MQTT broker, we can follow the below steps to add the gateways to this tool:

1. Add the MQTT broker in the ***Tool Settings page***, then go to this home page to choose the broker for a connection.

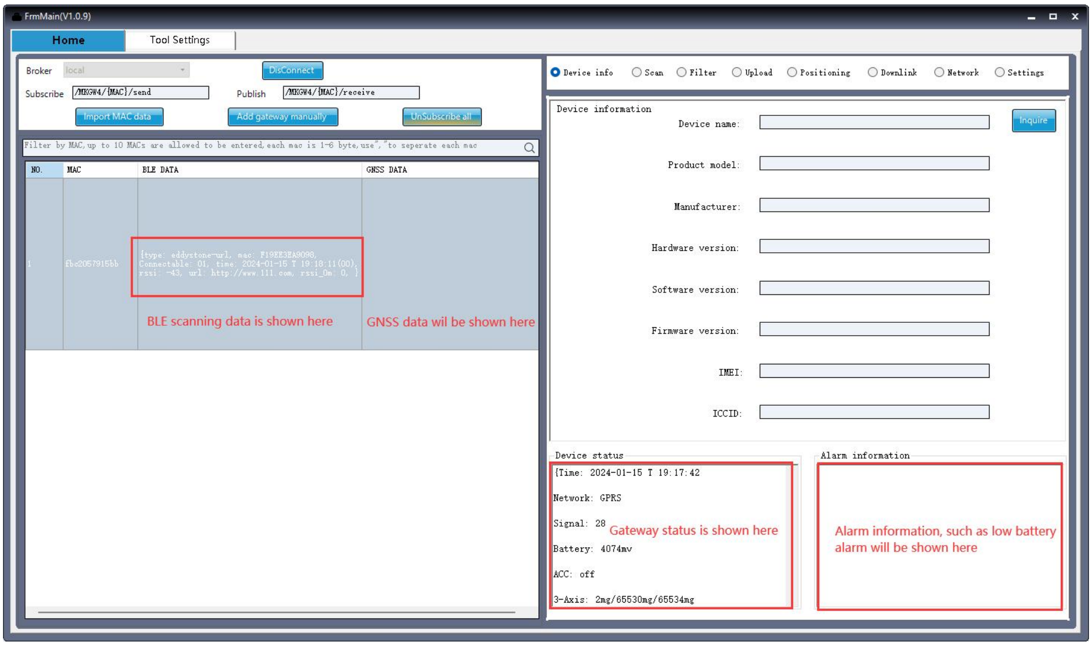
2. Click 

button and fill in the gateway mac address, publish topic and subscribe topic. After click confirm button, the gateway will be listed in the data table, and the tool will start monitoring the gateway data flow.



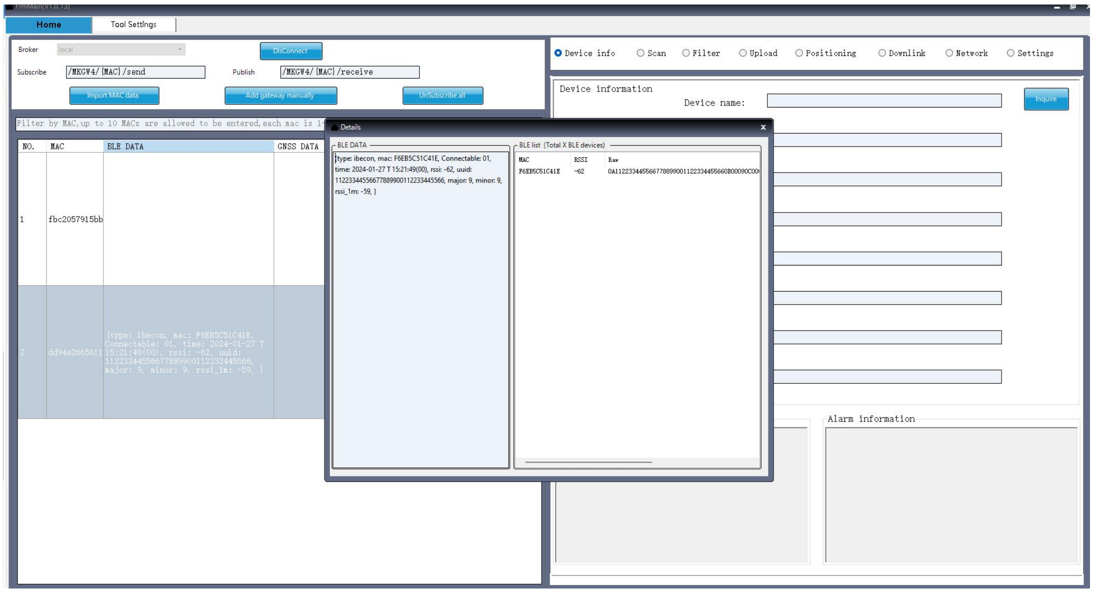
4.2 View gateway data

Once it receives the gateway reporting data, it will display the data in the corresponding columns. This tool will decode the HEX raw data into detailed BLE tag information, GNSS information, status information and alarm information.



Click the gateway row and click right key, we can do some operations on the gateway, such as reboot, reset, power off, inquire location and unsubscribe.

Click the gateway row and double click the BLE data column, we can see the complete data and total BLE devices captured by the gateway.



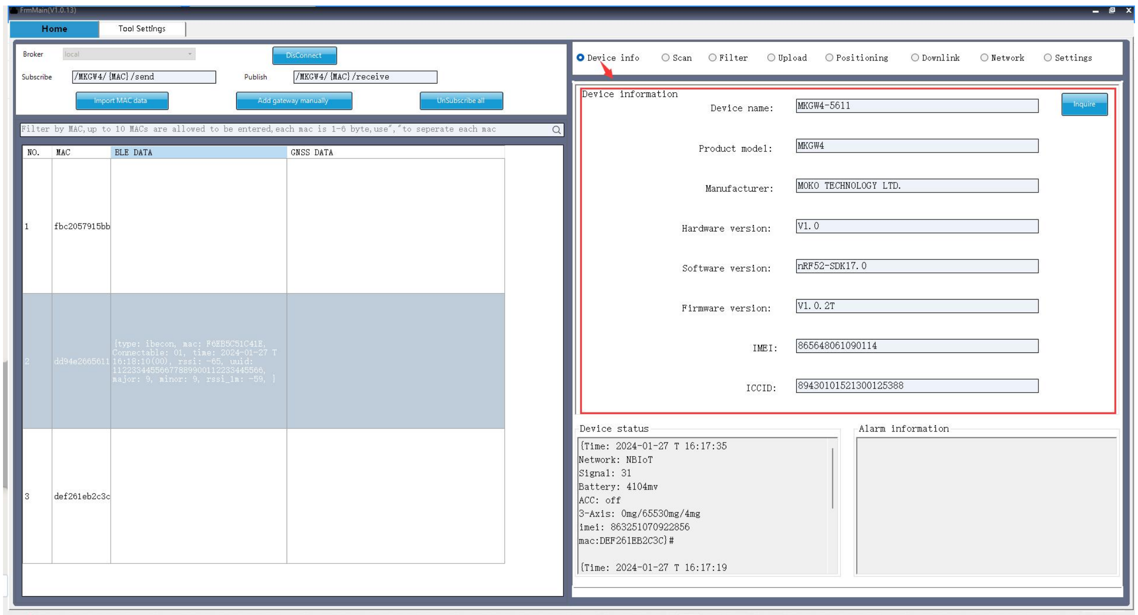
4.3 Remote configurations

To remotely configure and manage the gateway. For each parameter, there is a inquire button and set button which can be used to get the current value and set a new value to the gateway.
For more details to set the parameters please refer to the configuration APP manual.

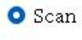
4.3.1 Device info

Click Device info to inquire the basic information of the gateway, including the name, model, firmware version, IMEI and SIM card ICCID.

Select a gateway by clicking the table row and click the inquire button, if the gateway is online, the tool will get reply from gateway show the information in the corresponding boxes.



4.3.2 Scan& report mode

Click the 

button to inquire and configure the scan&report mode and related parameters of the gateway.

There are 5 modes can be selected:

- **Real time scan& immediately report**

The gateway BLE scan is always on, it immediately transfers BLE data to cloud through the cellular network. This mode is mainly used for the external power supply, the gateway is always connected with cloud to ensure the immediate data reporting.

- **Real time scan& periodic report**

This is the default work mode. BLE scan is always on, the gateway reports BLE data according to the preset interval. In most time, the gateway cellular module is in sleep mode, when the reporting time is coming, the gateway cellular exits sleep mode and connect to network, then transfers BLE data to cloud. The gateway cellular module will disconnect from network and enters sleep mode after data reporting is finished

- **periodic scan& immediately report**

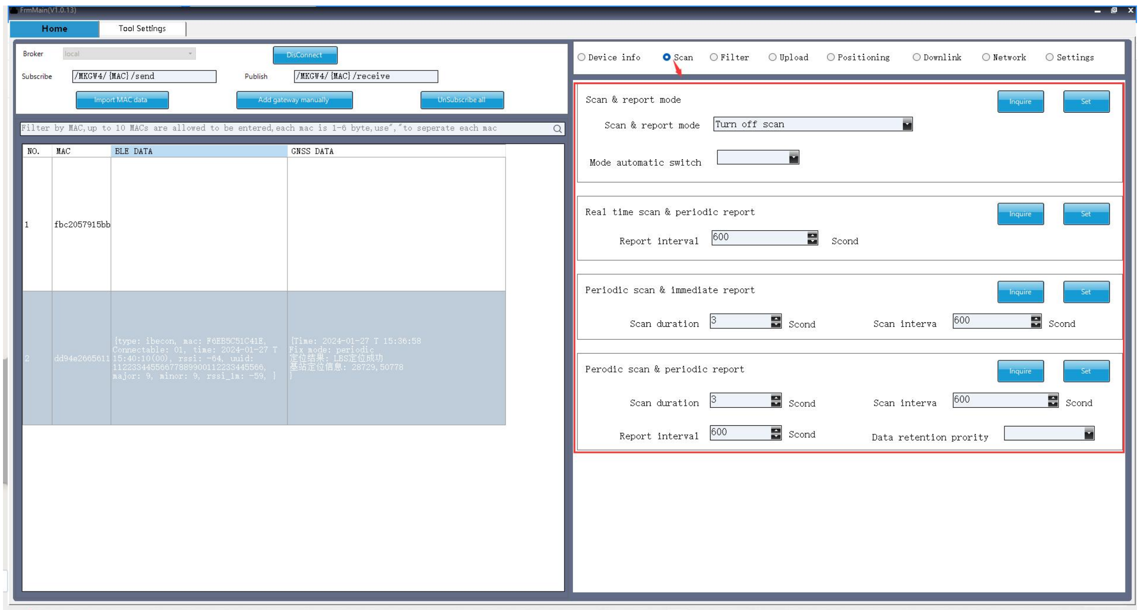
BLE scan is turned on periodically and keep scanning for a preset time length. In most time, the gateway cellular module is in sleep mode, When the scan time is coming, the gateway exits sleep mode, connects to network and turns on BLE scanning. BLE scanning will be stopped after the scan time expires, the cellular module will disconnect from network and enters sleep mode after data reporting is finished.

- **periodic scan& periodic report**

BLE scanning and data reporting are both periodical, the two periods can be different. The gateway will report the BLE data from the last reporting time, it may includes more than one rounds of BLE scanning data.

- **Turn off scan**

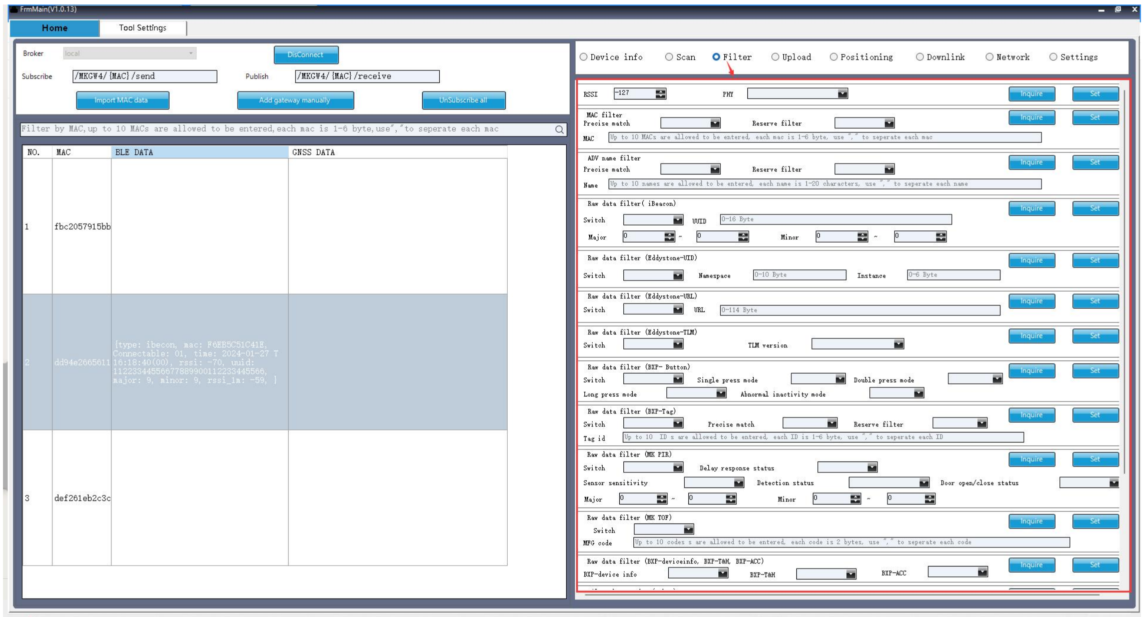
The gateway BLE scan is turned off, there will be no any BLE data reporting.



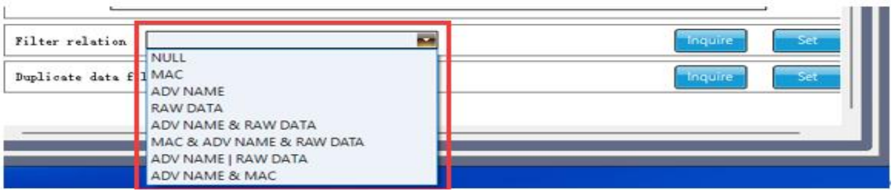
4.3.3 Filter

Click the Filter button to inquire and configure the BLE scan filtration of the gateway.

The gateway supports filter BLE scan data by RSSI, MAC, ADV name, raw data types (includes iBeacon, Eddystone-UID, Eddystone-URL, Eddystone-TLM, BXP-button, BXP-Tag, MK PIR, MK TOF). The process logic among different filtration types are also configurable.



Configure process logic for filtration type:

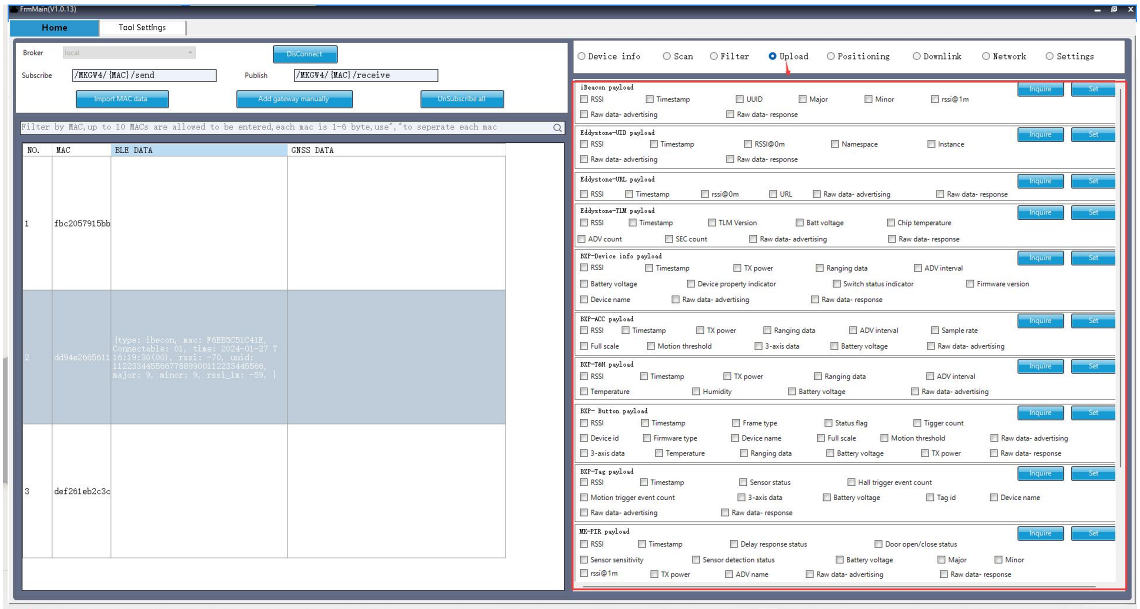


Configure the judgement rule for duplicate data filter:

4.3.4 Upload

Click the Upload button to inquire and configure the uploading payload for BLE data.

Select the upload option for each Beacon frame type. The checked items will be uploaded.

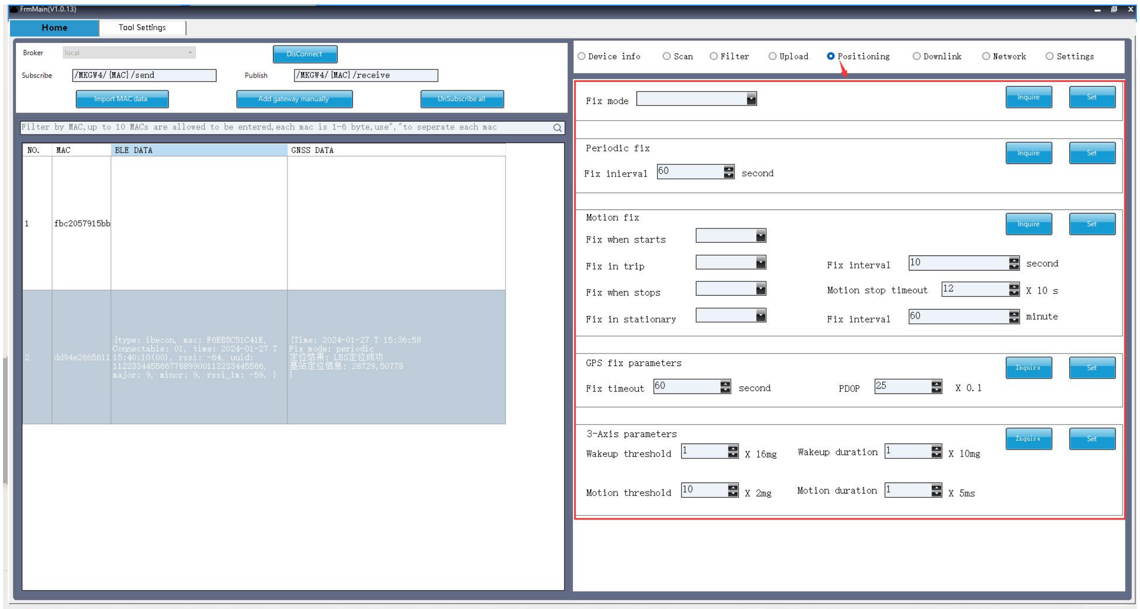


4.3.5 Positioning

Click the Positioning button to inquire and configure the positioning mode and related parameters.

There are 3 fix modes for selection:

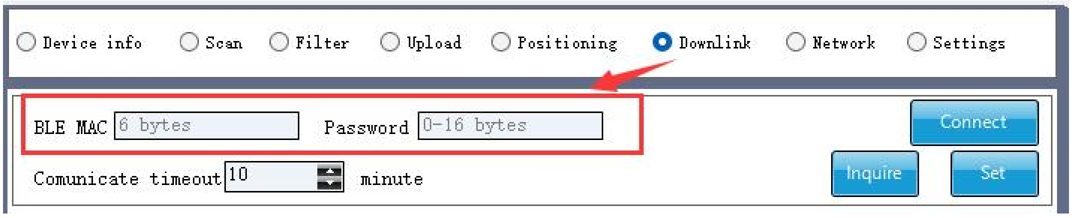
- Periodic fix: The gateway fixes and report location periodically.
- Motion fix: The gateway fix location based on the movement status. The fix interval in different movement status can be set separately.
- Turn off GPS: The gateway GPS function will be disabled.



4.3.6 Downlink

Click the Downlink to remotely connect and communicate with Bluetooth beacon. At present, only MK button can be connected. Input the MAC address and password of the Button device, click Connect button, the gateway will establish a connection with the Button.

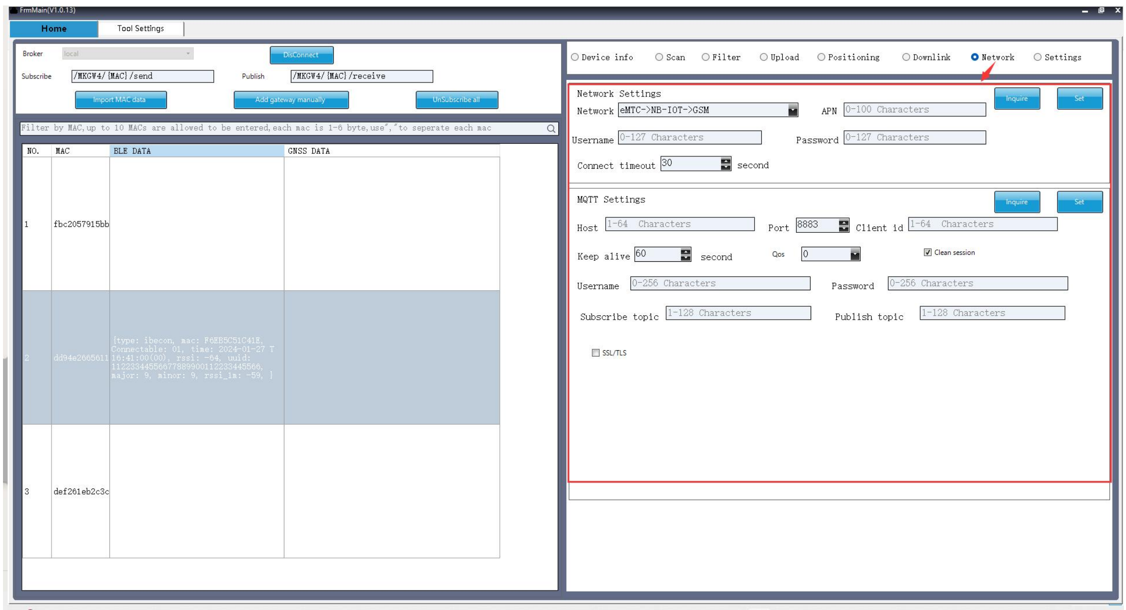
After a successful connection, we can inquire the basic device information from the Button device, such as firmware version, battery voltage and so on.
We can also remotely control the LED and buzzer of the Button via the gateway, dismiss the alarm status and remotely upgrade the firmware of the Button device.

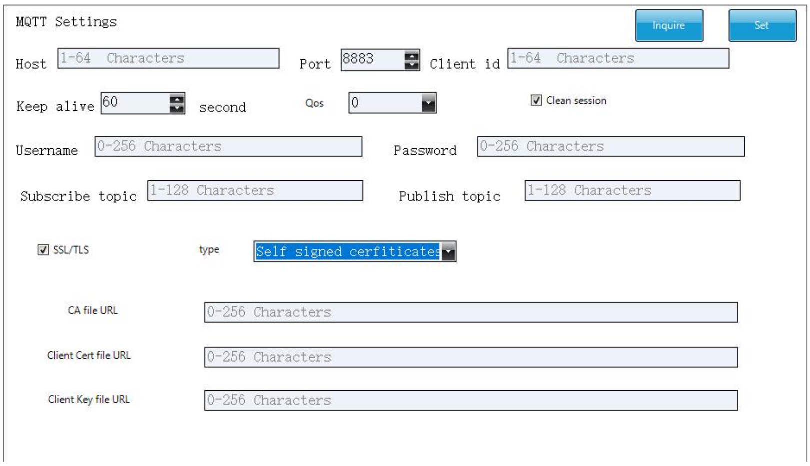


4.3.7 Network

Click the Network button to remotely configure the cellular network and MQTT settings.
If the MQTT broker is required SSL/TLS encryption, we need select the SSL/TLS option and upload correspond certificate files, the certificate file will be obtained via HFS server. The URL will be like:
http://47.104.172.169:8080/moko_cert/CA_cert

Once all changes are completed, click the set button, then the gateway will reboot immediately and apply the new network or MQTT configuration for re-connection.



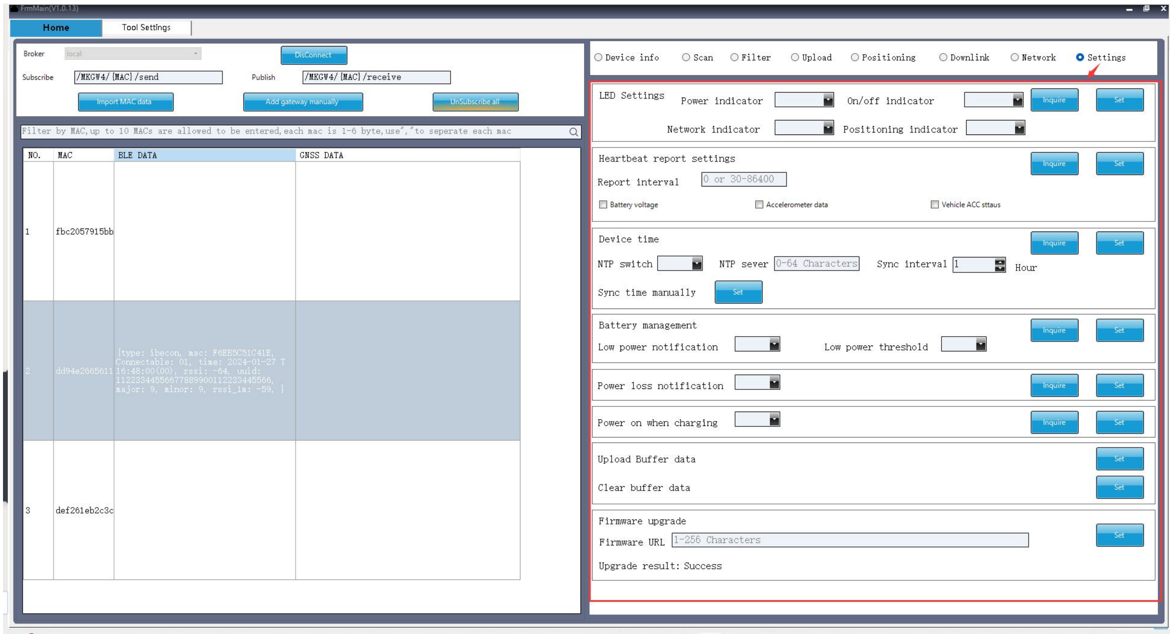


Note: The MQTT broker settings here should be consistent with the one in tool settings. And you should choose the same broker on homepage as well.

4.3.8 Setting

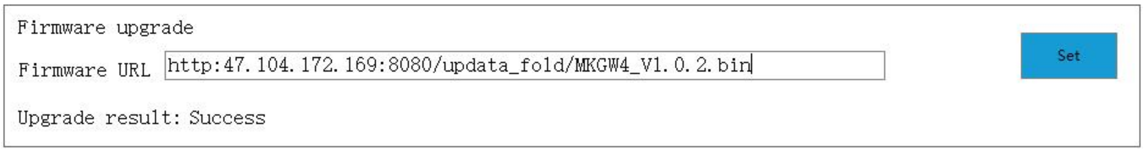
Click the Settings button to inquire and configure the system parameters of the gateway.

In this page, we can change the LED indicator status, set heartbeat interval, device time sync, battery low power notification, upload/clear buffer data and firmware upgrade.



The gateway will stores offline data in buffer, we can click the set button behind the upload buffer data, then the gateway will start to upload buffering data.
We can also click the set button behind the clear buffer data, then the gateway will delete all buffering data.

The last one is firmware upgrade. When run the firmware upgrading, the gateway will obtain firmware file from a HFS server. Input the correct firmware URL and click set button, the gateway will start upgrading. After the upgrade is done, the upgrade result will be prompted here.



Revision History

Revision	Description	Editor	Date
V1.0	Initial version	GuoJiasen	2024.1.15

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